FIITJEE **JEE (Advanced)-2018**

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST - III

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 231**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-C & Section-D**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. ***Section-A (01 – 03, 24 – 26, 47 – 49)*** *contains 9 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

   Section-A (04 – 08, 27 – 31, 50 – 54) *contains 15 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer.*

   Partial Marks **+1** for each correct option provided no incorrect options is selected.

   Section-A (09 – 10, 32 – 33, 55 – 56) *contains 3 paragraphs. Based upon paragraph, 2 multiple choice questions have to be answered. Each question has* ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer. There is no negative marking.*

2. ***Section-C (11 – 20, 34 – 43, 57 – 66)*** *contains 30 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

3. ***Section-D (21 – 23, 44 – 46, 67 – 69)*** *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

PART – I (Physics), PART – II (Chemistry), PART – III (Mathematics):

(SECTION – D)

For questions **21 to 23, 44 to 46, 67 to 69.**
Numerical answer type questions with answer XXXXX. XX

If answer is 348.4 / 251.37 / 213

Correct Method :

0	0	3	4	8	.	4	0
0	0	2	5	1	.	3	7
0	0	2	1	3	.	0	0

Wrong Method :

	3	4	8		.	4	
3	4	8			.		4
		3	4	8	.		4
	3		4	8	.	4	
	2		5	1	.	3	7
		2	1	3	.		
		2	1	3	.	0	
		2	1	3	.		0
		3	4	8	.	4	0
		2	5	1	.	3	7
		2	1	3	.	0	0

Physics

PART – I

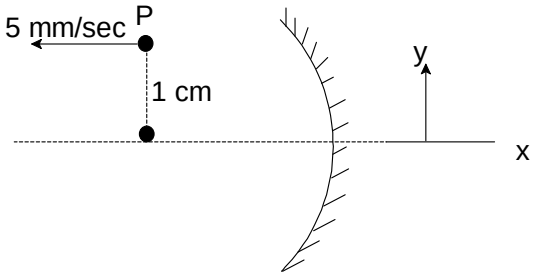
SECTION – A
(One Options Correct Type)

This section contains **3 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **ONLY ONE** option is correct.

1. The internal energy of a gas is given by U = 2PV. It expands from V_0 to $2V_0$ against a constant pressure P_0. The heat absorbed by the gas in the process is
(A) $2P_0V_0$ (B) $4P_0V_0$
(C) $3P_0V_0$ (D) P_0V_0

2. A metal rod of length ℓ & area of cross-section A is suspended to the celling by one of its end. The density of metal is ρ, young modulus of elasticity is Y. Choose the correct option
(A) The stress at mid point is $\ell\rho g$
(B) Extension in rod due to it's own weight $\Delta\ell = \frac{\rho g \ell^2}{2Y}$
(C) potential energy stored in rod due to its own weight $U = \frac{\rho^2 AL^3}{3Y}$
(D) The stress at mid point is $2\ell\rho g$

3. A point object P is moving towards left with speed 5 mm/sec parallel to optical axis of a concave mirror of focal length f = 20 cm. The vertical separation between object and optical axis is 1 cm. Find velocity of image of object in vector form, when foot of perpendicular from object on the optical axis is at a distance 30 cm from pole.



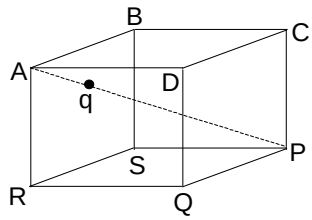
(A) $\vec{V_i} = 20\hat{i} + 4\hat{j}$ mm / sec
(B) $\vec{V_i} = 20\hat{i} + \hat{j}$ mm / sec
(C) $\vec{V_i} = 20\hat{i} - \hat{j}$ mm / sec
(D) $\vec{V_i} = 20\hat{i}$ mm / sec

Space for Rough work

(One or More than one correct type)

This section contains **FIVE** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

4. A charge 'q' is placed on the diagonal AP of a cube at a distance $\frac{AP}{3}$ from the point A. Choose the correct options.



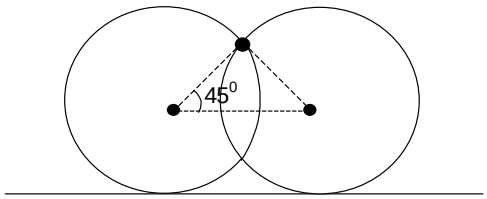
(A) The sum of electric flux passing through the surface ABCD and PQRS is $\frac{q}{3\varepsilon_0}$

(B) The sum of electric flux passing through the surface ABCD and PQRS is $\frac{q}{8\varepsilon_0}$

(C) The flux through both the surfaces ABCD and PQRS are same

(D) The flux through the surface ABCD is larger than the flux through surface PQRS

5. Two identical uniform large rings each of mass m are connected through a bead of same mass, which can move freely. When bead is released it starts sliding down. The large rings roll apart over a sufficiently rough horizontal surface. Whole system is released from rest as shown in figure.



Choose the correct Option(s). (Neglect friction between bead and large rings)

(A) Acceleration of the bead at the initial moments is $\frac{g}{9}$

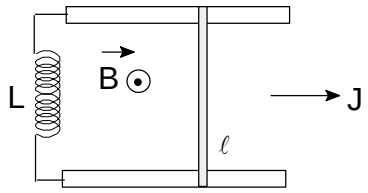
(B) Acceleration of the bead at the initial moment is $\frac{g}{5}$

(C) At the initial moment normal contact between bead and any of the ring is $\frac{4\sqrt{2}mg}{9}$

(D) Normal contact between bead and any of the ring is $\frac{2\sqrt{2}mg}{5}$

Space for Rough work

6. Two parallel resistance less rails are connected by an inductor of inductance L at one end as shown in the figure. A magnetic field B exists in the space which is perpendicular to the plane of the rails. Now a conductor of length ℓ, mass m and negligible resistance is placed transverse on the rails and given an impulse J towards the rightward direction. Then choose the correct option(s).



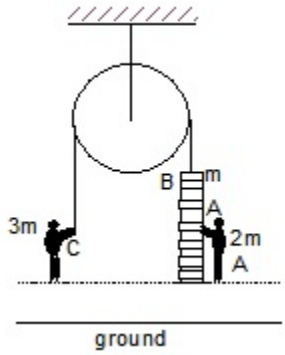
(A) Velocity of the conductor is half of the initial velocity after a displacement of the conductor $d = \sqrt{\frac{3J^2L}{4B^2\ell^2m}}$

(B) Current flowing through the inductor at the instant when velocity of the conductor is half of the initial velocity is $i = \sqrt{\frac{3J^2}{4Lm}}$

(C) Velocity of the conductor is half of the initial velocity after a displacement of the conductor $d = \sqrt{\frac{3J^2L}{B^2\ell^2m}}$

(D) Current flowing through the inductor at the instant when velocity of the conductor is half of the initial velocity is $i = \sqrt{\frac{3J^2}{Lm}}$

7. Man A of mass 2m is standing on a ladder of mass m. Ladder is attached with an inextensible light string and man C of mass 3m is holding the other end of string. Initially, the whole system is at rest. Assume whole system is at sufficient height and pulley is frictionless. Also A and C are at same horizontal level. Then choose the correct options. [g = 10 m/s^2].



(A) If man A starts climbing up on the ladder with a constant acceleration of 4 m/s^2 w.r.t ground then, the vertical distance between A and C after 1 sec is 1 m

(B) If man C starts climbing up on the rope (while A remain at rest with respect to ladder) with constant acceleration of 4 m/s^2 with respect to ground then, the vertical distance between A and C after 1 sec is 2m

(C) If both man A and C starts climbing up with constant acceleration 4 m/s^2 with respect to ground then, acceleration of ladder is 4 m/s^2 upward

(D) If both man A and C starts climbing up with constant acceleration 4 m/s^2 with respect to ground then, acceleration of ladder is 3 m/s^2 upward

Space for Rough work

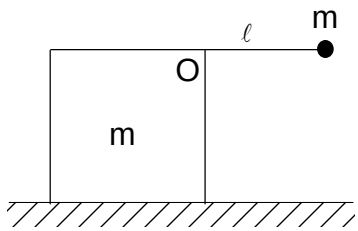
8. In the Searle's experiment, after every step of loading , why should we wait for two minutes before taking the readings?
(A) So that the wire can have its desired change in length.
(B) So that the wire can attain room temperature.
(C) So that vertical oscillations can get subsided.
(D) So that the wire has no change in its radius.

(Paragraph Type)

This section contains **ONE** paragraph. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 9 and 10

A particle of mass m is connected to a block of same mass by an inextensible string of length ℓ. The block is free to slide on the smooth horizontal surface. Size of the block is large enough so, that is doesn't topple and particle doesn't hit on the ground. Now the particle is released from the horizontal position of the string as shown in the figure. Answer the following two questions.



9. Speed of the block as function of θ (where θ is the angle made by the string with downward vertical) will be:
(A) $\sqrt{\frac{g\ell\cos\theta}{1+\tan^2\theta}}$ (B) $\sqrt{\frac{2g\ell\cos\theta}{1+\tan^2\theta}}$
(C) $V=\sqrt{\frac{g\ell\cos\theta}{1+2\tan^2\theta}}$ (D) $V=\sqrt{\frac{g\ell\cos\theta}{2+\tan^2\theta}}$

10. Normal reaction offered by ground onto the wedge as a function of θ. (where θ is the angle made by the string with downward vertical) will be:
(A) $mg\cos\theta\left(\frac{5+\sin^2\theta}{\left(1+\sin^2\theta\right)^2}\right)$ (B) $mg\left(1+\frac{\cos^2\theta\left(5+\sin^2\theta\right)}{\left(1+\sin^2\theta\right)^2}\right)$
(C) $mg\cos\theta\left(\frac{5+\sin^2\theta}{1+\sin^2\theta}\right)$ (D) $mg\left(1+\frac{\cos^2\theta\left(3+\sin^2\theta\right)}{\left(1+\sin^2\theta\right)^2}\right)$

Space for Rough work

SECTION – C
(Single digit integer type)

This section contains **TEN** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

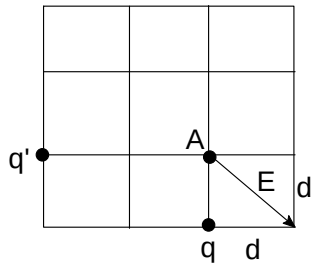
11. An ant with mass m is standing peacefully on top of a horizontal stretched rope. The rope has mass per unit length μ and is under tension F. Now starts a sinusoidal transverse wave of wavelength λ propagating along the rope. The motion of the rope is in a vertical plane. What minimum amplitude (in mm) will make the ant feel weight less momentarily? Assume that m is so small that the pressure of the ant has not effect on the propagation of the wave. (Given $\lambda = 0.5\text{m}$, $\mu = 0.1\,\text{kg/m}, F = 3.125\text{N}, \text{Take} g = \pi^2$)

12. A photon strikes a hydrogen atom in its ground state to eject the electron with kinetic with energy 16.4 eV. If 25% of the photon energy is taken up by the electron, the energy of the incident photon is $(24 \times X)$ eV then 'X' is;

13. Two men A and B, A standing at periphery of circle of radius R and B standing at centre of the circle. A starts moving on the periphery with constant acceleration "a" and B starts moving with constant velocity v at an angle θ, from the line joining initial position of A and centre, such that they meet during subsequent motion, for this purpose maximum possible value of v is $\sqrt{\frac{aR}{\alpha\theta}}$. Here α is an integer. Find α.

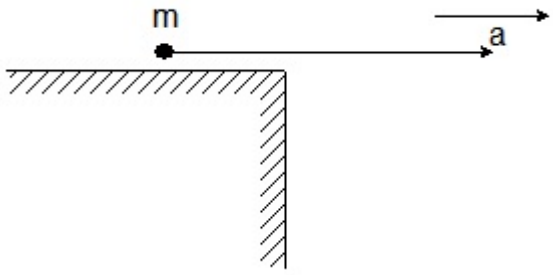
14. The range of voltage of a galvanometer used as a voltmeter can be changed to n times its original value with the help of a 27Ω resistance. Using the same galvanometer as an ammeter, its range of current can be changed to n times its original value using a 3Ω shunt. Find the internal resistance of the meter $(\text{in } \Omega)$.

15. In given arrangement side of each small square is d. Two point charges q and q' are placed as shown in the figure. If the total electric field at A is as shown in the figure below and potential at point A is given by $\frac{-nq}{8\pi\varepsilon_0 d}$ then find n.

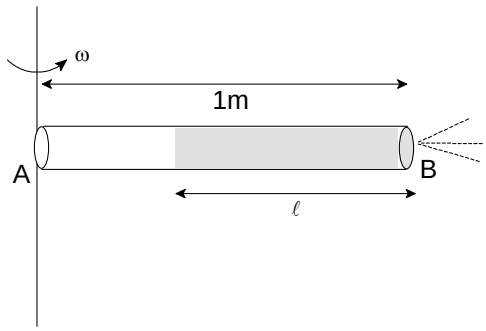


Space for Rough work

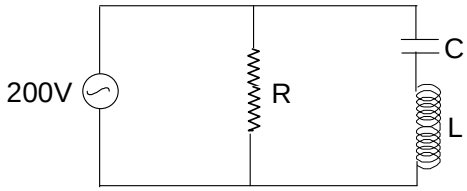
16. A small particle is attached to an inextensible string and placed on a table. The other end is pulled with constant acceleration $\frac{4g}{3}$ towards right. Initially the particle was at rest. After the particle leaves the edge of table, the maximum vertical displacement of the particle is from initial horizontal level is $\frac{48}{25}$m. Find the length of string in m.



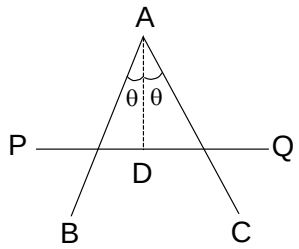
17. A narrow horizontal tube AB of length 1m open at A and closed at B is filled with an ideal fluid. The end B has a small orifice. The tube is set in rotation in horizontal plane with angular velocity $\omega = 10\text{rad/sec}$. About a vertical axis passing through A. The efflux velocity of the fluid is V m/s. The length the fluid is $\ell = 0.4\text{m}$. Find the value of V.



18. In the circuit diagram shown, X_C = 100Ω, X_L = 200Ω & R = 100Ω. The effective current through the source is $\sqrt{X}$ ampere then find out value of X:

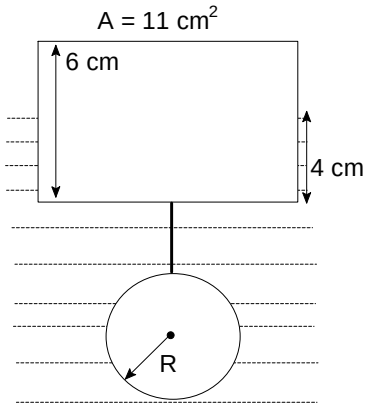


19. In the figure shown AB, AC and PQ are thin, smooth, rigid wires, AB and AC are joined at A and fixed in vertical plane. $\angle BAC = 2\theta = 90^0$ and line AD is angle bisector of angle BAC. A liquid of surface tension T = 0.025N/m forms a thin film in the triangle formed by intersection of the wires AB, AC and PQ. In the figure shown the uniform wire PQ of mass 1 gm is horizontal and in equilibrium under the action of surface tension and gravitational force. Find the time period of SHM of PQ in vertical plane for small displacement from its mean position. In the form $\frac{\pi}{X}$s and find value of X.



Space for Rough work

20. Figure shows a uniform metal ball suspended by thread of negligible mass from an upright cylinder that floats partially submerged in water. The cylinder has height 6 cm, face area 11 cm^2 on the top and bottom and density 0.5 g/cm^3. 4 cm of cylinder's height is inside the water surface. Density of the metal ball is $8gm/cm^3$. R is the radius of the ball. It is found that $R^3 = \frac{3}{\alpha} cm^3$, where α is an integer $\left(\rho_w = 1 gm/cm^3\right)$ (system is in equilibrium) [$\pi = 22/7$]. Find α



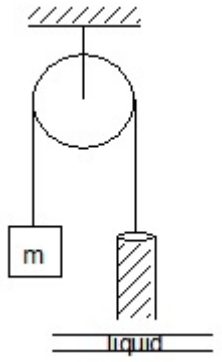
SECTION – D
(Numerical Based XXXXX.XX answer Type)

This section contains **3 questions**. Each question, when worked out will result in numerical answer Type with answer xxxxx.xx.

21. A radionuclide with disintegration constant λ is produced in a reactor at a constant rate $\alpha(=2\lambda)$ nuclei per second. At t = 0, there are no nuclei present in the reactor. During each decay energy E_0 is released. 20% of this energy is utilized in increasing the temperature of water. Find the increase in temperature of m mass of water in time $t = T_{1/2}$ if specific heat of water is s. Assume that there is no loss of energy through water surface. (Given E_0 = 100 times the energy required to raise the temperature of mass m of water from 0^0C to 100^0C)

22. A parachutist jumps from the top of a very high tower with a siren of frequency 800 Hz on his back. Assume his initial velocity to be zero. After falling freely for 12 s, he observes that the frequency of sound heard by him reflected from level ground below him is differing by 700 Hz w.r.t. the original frequency. What was the height of tower. Velocity of sound in air is 330 m/s, and $g = 10 m/s^2$.

23. A uniform cylinder of mass 2m and a block of mass m are connected by a light string passing over a fixed smooth pulley as shown in the figure. The system is released from rest when the lower edge of the cylinder just touches the surface of the liquid. The area of cross-section of the cylinder is A and its length is L. What should be the density of the liquid, if the cylinder stops at the instant when the upper edge of the cylinder just touches the surface of the liquid? Assume infinite extension of the liquid and the viscosity of the liquid to be zero. $\left(\text{Given } m = 1 kg,\ A = 10^{-2} m^2,\ L = 1m\right)$



Space for Rough work

Chemistry

PART – II

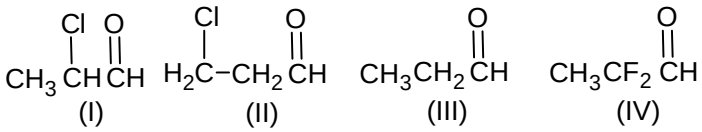
SECTION – A
(One Options Correct Type)

This section contains **3 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **ONLY ONE** option is correct.

24. Which of the following is not a balanced chemical reaction?
(A) $2MnO_4^- + H_2O_2 + 6H^+ \longrightarrow 2Mn^{+2} + 3O_2 + 4H_2O$
(B) $3P_4S_3 + 38NO_3^- + 8H_2O + 20H^+ \rightarrow 12H_3PO_4 + 9SO_4^{-2} + 38NO$
(C) $5SO_3^{-2} + 2MnO_4^- + 6H^+ \rightarrow 5SO_4^{-2} + 2Mn^{+2} + 3H_2O$
(D) None of these

25. Which of the following is not a correct statement about $Al_2(CH_3)_6$?
(A) Maximum no. of atoms present in same plane = 10
(B) Maximum no. of C - atoms present in same plane = 4
(C) Maximum no. of atoms present in a plane which contain two C atom = 6
(D) None of these

26. Arrange these compounds in orders of increasing equilibrium constant for hydrate formation.



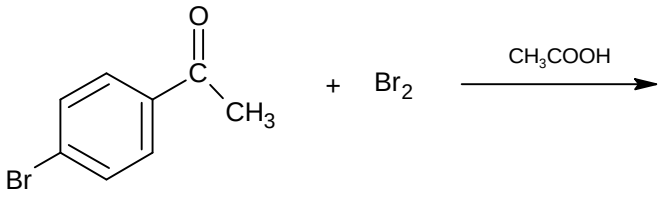
(I) (II) (III) (IV)
(A) I < II < III < IV
(B) III < II < I < IV
(C) IV < I < II < III
(D) None of these

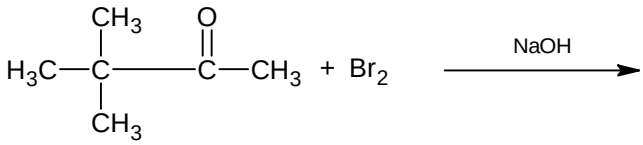
Space for Rough work

(One or More than one correct type)

This section contains **FIVE** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

27.

(I) 

(II) 

Choose the incorrect statement(s) for mono halogenation.
(A) I is more suitable than II
(B) II is more suitable than I
(C) I and II are equally suitable
(D) None of two are suitable for mono halogenation

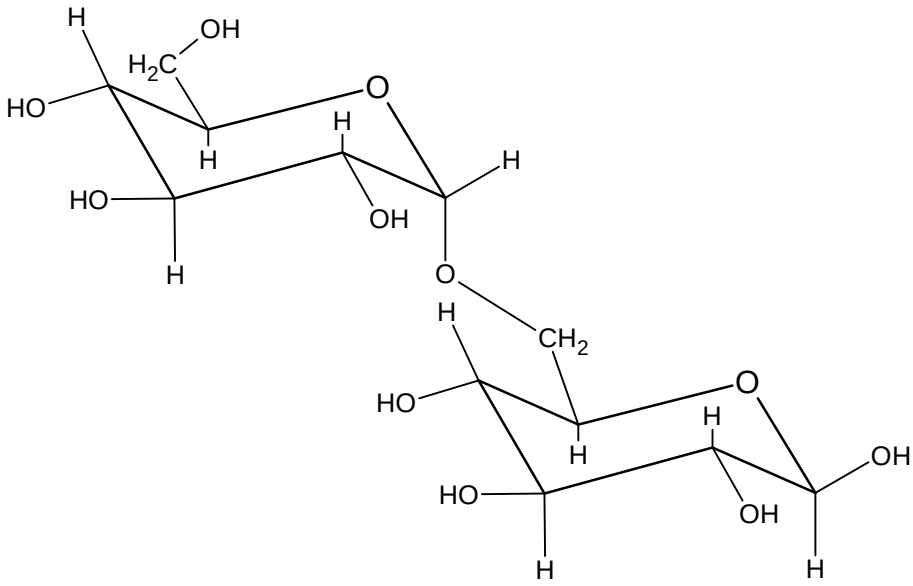
Space for Rough work

28. Which of the following is/are reducing sugar(s)?

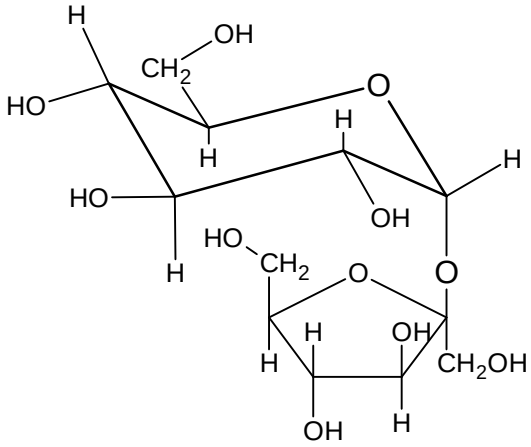
(A) Methyl $\alpha - D -$ galactopyranoside

(B) $\beta - L -$ idopyranose (an aldohexose)

(C)



(D)

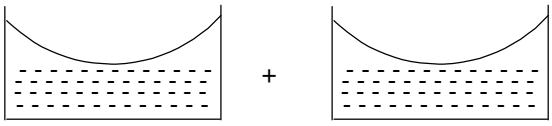


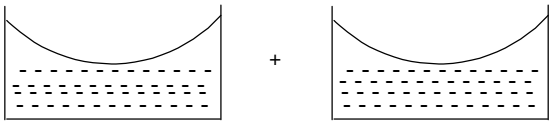
Space for Rough work

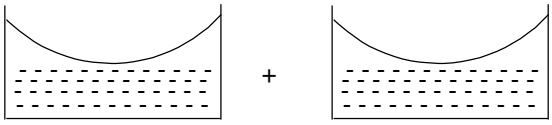
29. $[Cr(CN)_6]^{3-}$ $[Mn(CN)_6]^{3-}$ $[Fe(CN)_6]^{3-}$ $[Co(CN)_6]^{3-}$
I II III IV

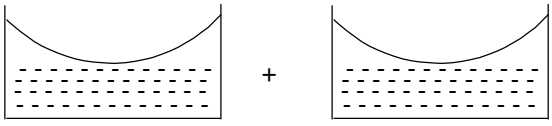
Choose the correct statements.
(A) I < II < III < IV (strength of metal carbon bond)
(B) I > II > III > IV (strength of metal carbon bond)
(C) I < II < III < IV (strength of carbon nitrogen bond)
(D) I > II > III > IV (strength of carbon nitrogen bond)

30. Which combinations behave as buffer solution?

(A)  1 Lit 1 M NH_3 + 1 Lit 1 M CH_3COOH

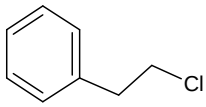
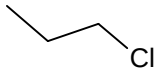
(B)  1 1 M NaHCO + 1 1 M Na_2CO

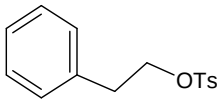
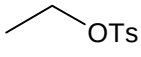
(C)  1 Lit 1 M HCl + 1 Lit 2 M CH_3COONa

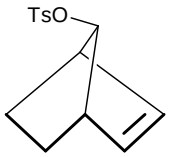
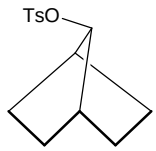
(D)  1 Lit 1 M NaOH + 1 Lit 0.5 M NH_4Cl

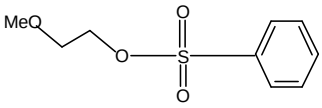
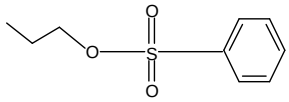
Space for Rough work

31. Choose the correct statements:

(A)  reacts with water faster than 

(B)  reacts with CF_3COOH faster than 

(C)  reacts with acetic acid faster than 

(D)  reacts with AcOH faster than 

(Paragraph Type)

This section contains **ONE** paragraph. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 32 and 33

Since boron has three valence-shell electrons and nitrogen has five, the molecule BN is isoelectronic with C_2. Also some of the allotropic forms that exist for carbon also exist for compounds that have the formula $(BN)_x$

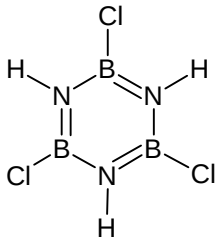
32. Choose the incorrect statement.
(A) borazon is one of the hardest compound known
(B) cubic form of $(BN)_x$ is called borazon
(C) $B_3N_3H_6$ is called borazon
(D) graphite form of $(BN)_x$ is used as a high temperature lubricant

Space for Rough work

33. Choose the incorrect statement.

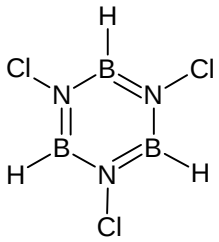
(A) on large scale the reaction of $(NH_2)_2CO$ with $B(OH)_3$ at high temperature in an atmosphere of ammonia is used to produce borazine

(B) Trichloroborazine ($B_3N_3Cl_3H_3$) has the structure



(C) reaction of $B_3N_3Cl_3H_3$ with $LiAlH_4$ gives borazine

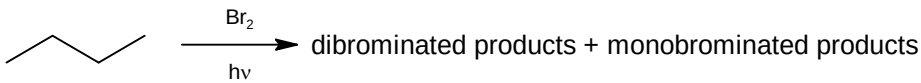
(D) trichloroborazine ($B_3N_3Cl_3H_3$) has the structure



SECTION – C
(Single digit integer type)

This section contains **TEN** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

34.

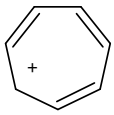


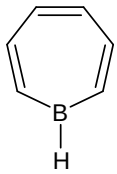
No of optically active products are ….

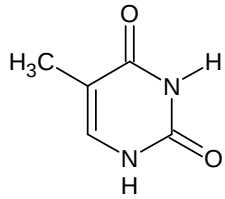
Space for Rough work

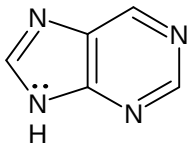
35. How many compounds/ions are aromatic?

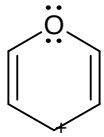
(A) Borazine (B) Graphite (C) Borazon

(D) 

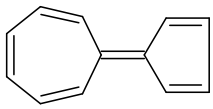
(E) 

(F) 

(G) 

(H) 

(I) 

(J) 

36. Number of carbohydrats which show mutarotation are …..

(A) α - D - (+) – glucopyranose (B) β - D – (+) – glucopyranose
(C) α - D – (–) – fructofuranose (D) β - D – (–) – fructofuranose
(E) sucrose (F) maltose
(G) lactose

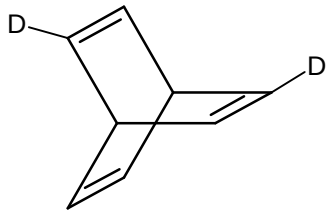
37. Number of compounds which show more basic character than NH_3 (in aqueous condition) are …

(A) $N(CH_3)_3$ (B) $NH(CH_3)_2$ (C) NH_2CH_3
(D) $N(C_2H_5)_3$ (E) $NH(C_2H_5)_2$ (F) $NH_2C_2H_5$
(G) $N(n\text{-}Pr)_3$ (H) $NH(n\text{-}Pr)_2$ (I) $NH_2(n\text{-}Pr)$
(where n-Pr = normal-propyl)

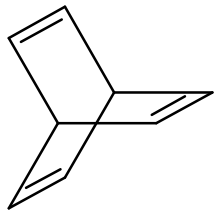
Space for Rough work

38. Number of compounds which show optical activity are …..

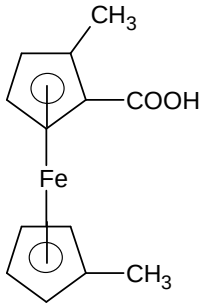
(A)



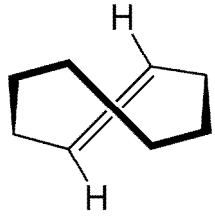
(B)



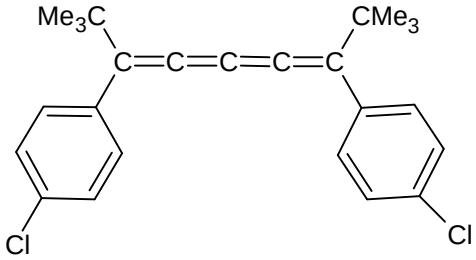
(C)



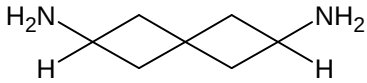
(D)



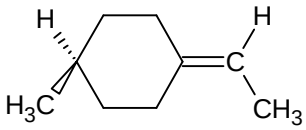
(E)



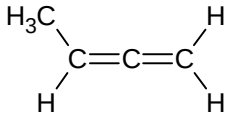
(F)



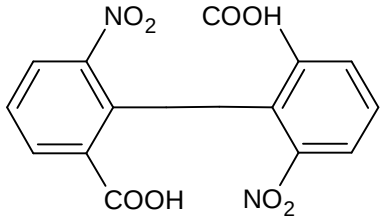
(G)



(H)



(I)



Space for Rough work

39. The percentage of Ni^{+2} and Ni^{+3} ions in $Ni_{0.98}O_{1.00}$ is x and y respectively then value of $\frac{x}{4y}$ is ....

40. An aqueous solution of glucose boils at 100.01°C. The molar elevation constant for water is 0.5 K kg mol^{-1}. If number of glucose molecule in the solution containing 100 g water is 1.2×10^x, then $\frac{x}{3}$ is

41. Find number of compound(s) in which atoms of anion are bonded by only one covalent bond.
NaOH, Na_2O, Na_2O_2, BaO_2, CaC_2, CaH_2, $BaCl_2$, Na_2S_2

42. Find number of yellow ppt from given below:
AgCl, AgBr, $PbCrO_4$, $Fe(SCN)_3$, Hg_2I_2, $PbBr_2$, PbI_2, $Fe(OH)_2$

43. Find number of order which is/are correctly arranged.
(I) $CH_4 < SiH_4 < SnH_4$ (boiling point)
(II) $SbH_3 > NH_3 > PH_3$ (boiling point)
(III) $H_2O > SeH_2 > H_2S$ (boiling point)
(IV) $HF > HBr > HCl$ (boiling point)
(V) $F_2 < Cl_2 < Br_2 < I_2$ (bond length)
(VI) $F_2 > Cl_2 > Br_2 > I_2$ (bond energy)
(VII) chair form of cyclohexane > half chair form of cyclohexane (stability)

Space for Rough work

SECTION – D
(Numerical Based XXXXX.XX answer Type)

This section contains **3 questions**. Each question, when worked out will result in numerical answer Type with answer xxxxx.xx.

44. Two moles of an ideal gas were heated in a rigid container such that its internal energy rises by (40 $\times$ 8.314) J and temperature increase by 8 K. One mole of the same gas undergoes a reversible adiabatic expansion such that the ratio of final volume to initial volume is 32. If $\log\left(\frac{P_1}{P_2}\right) = x\log 2$, where P_1 & P_2 are initial & final pressure respectively then x is
(R = 8.314 J mol^{-1} K^{-1})

45. The following equilibrium was established at constant T in a rigid container starting with only AB(g).
$AB(g) \rightleftharpoons A(g) + B(g)$
The degree of dissociation of AB is 20% under these conditions & the relative vapour density of the equilibrium mixture is 'y'. Neon gas (inert) is added in the container such that its mole fraction is 0.4. The vapour density of the gaseous mixture in the container is 'z'. Then (y + z) is
(molar mass of AB(g) = 60 & neon = 20)

46. Out of the following wave-functions, only one is of the 2s orbital of H-atom

(a) $\psi = \left(\frac{1}{32\pi a_0^3}\right)^{\frac{1}{2}}\left(2-\frac{r}{a_0}\right)e^{-\frac{r}{2a_0}}$

(b) $\psi = \left(\frac{1}{24\pi a_0^3}\right)^{\frac{1}{2}}\left(6-\frac{6r}{a_0}+\frac{r^2}{a_0^2}\right)\left(e^{-\frac{r}{2a_0}}\right)$

(c) $\psi = \left(\frac{1}{32\pi a_0^3}\right)^{\frac{1}{2}}\left(2-\frac{r}{a_0}\right)e^{-\frac{r}{2a_0}}.\cos\theta$

(d) $\psi = \left(\frac{1}{32\pi a_0^3}\right)^{\frac{1}{2}}\left(\frac{r}{a_0}\right)e^{-\frac{r}{2a_0}}$

Using the correct wave-function, calculate the probability of find a 2s – electron in H atom on the spherical surface at a distance of twice the bohr's radius from the nucleus (a_0 = bohr's radius).

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **3 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **ONLY ONE** option is correct.

47. There are 20 locks and their 20 corresponding keys. If each lock has exactly 1 matching key then the maximum number of trials to match the locks with their corresponding keys
(A) 20! (B) 19!
(C) 1 (D) none of these

48. If a Δ ABC is formed by the lines $2x + y - 3 = 0$; $x - y + 5 = 0$ and $3x - y + 1 = 0$, then the cubic equation whose roots are the tangent of the interior angles of the triangle is
(A) $2x^3 + 3x^2 - 8x + 3 = 0$ (B) $2x^3 + 3x^2 + 8x + 3 = 0$
(C) $3x^3 + 3x^2 - 8x + 3 = 0$ (D) $3x^3 + 3x^2 + 8x + 3 = 0$

49. In ΔABC, $2x + 3y + 1 = 0$, $x + y = 1$ and $2x - y + 1 = 0$ are median through vertex A, altitude through vertex B and angle bisector of C respectively
(A) intercept made by BC on y-axis is $\frac{379}{59}$ (B) area of ΔABC is $\frac{401}{59}$
(C) intercept made by BC on x-axis is $\frac{379}{423}$ (D) none of these

Space for Rough work

(One or More than one correct type)

This section contains **FIVE** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

50. Let determinants $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & b_3 \end{vmatrix}$, $\Delta_x = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}$, $\Delta_y = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}$ and $\Delta_z = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$ and consider the statements

(i) if $\Delta = \Delta_x = \Delta_y = \Delta_z = 0$ then the equations $a_i x + b_i y + c_i z = d_i \ \ (i = 1,2,3)$ have infinitely many solutions

(ii) if $\Delta_x = \Delta_y = \Delta_z = 0 \neq \Delta$ then the equations $a_i x + b_i y + c_i z = d_i \ \ (i = 1,2,3)$ have unique solution which of the following are wrong

(A) (i) and (ii) are always are wrong
(B) (i) may or may not be true but (ii) always true
(C) in both (i) & (ii) equations are always consistent
(D) (i) is false & (ii) is true.

51. Let $1.1.2 + 2.2.3 + 3.4.4 + 4.8.5 + 5.16.6 + ...n \text{ terms} = \left(an^2 - bn + 2\right)2^n - c, \left(a,b,c \in N\right)$ then

(A) a + b = c
(B) 3a – b = c
(C) a + b + c = 4
(D) None of these

52. The extremum values of the function $f(x) = \dfrac{1}{\sin x + 4} + \dfrac{1}{\cos x - 4}$, where $x \in R$ is :

(A) $\dfrac{4}{8 - \sqrt{2}}$
(B) $\dfrac{2\sqrt{2}}{8 - \sqrt{2}}$
(C) $\dfrac{2\sqrt{2}}{4\sqrt{2} + 1}$
(D) $\dfrac{4\sqrt{2}}{8 + \sqrt{2}}$

Space for Rough work

53. If roots of $f(x) = x^2 - bx + c = 0$, b is an odd integer & $c \in z$; are prime numbers, then
(A) c is an odd integer (B) c is an even integer
(C) $2b - c = 4$ (D) $2b + c = 4$

54. Consider 10 sided convex polygon in which no two diagonals are parallel. 'S' is the set of distinct points of intersections of diagonals. A point is randomly picked up from 'S'. Which of the following is/are true.

(A) maximum value of probability that the point is outside the polygon is $\frac{35}{119}$

(B) maximum value of probability that the point is on the polygon is $\frac{2}{79}$

(C) maximum value of probability that the point is inside the polygon is $\frac{42}{79}$

(D) none of these

(Paragraph Type)

This section contains **ONE** paragraph. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Paragraph for Question Nos. 55 and 56

Let g be a continuous function on R and satisfies $g(x) + 2\int_0^{\frac{\pi}{2}} \cos x \sin tg(t)dt = \cos x$

55. The value of $g'(\frac{\pi}{6})$ is equal to

(A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$

(C) $-\frac{1}{4}$ (D) $\frac{\sqrt{3}}{2}$

Space for Rough work

56. The value of $\lim_{x\to 0}\frac{g'(x)}{xg(x)}$ is equal to

(A) 1 (B) $-\frac{1}{4}$

(C) 2 (D) -2

SECTION – C
(Single digit integer type)

This section contains **TEN** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

57. Let p be a positive integer and g(x) be a polynomial with integer coefficients satisfying $2\int_1^x g(t)\,dt + x^p = x\,g(x),\ where\ x \geq 1$. The sum of all possible values of p is.

58. Suppose p_1 & p_2 are the point of maximum and the point of minimum respectively of the function $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$ respectively, then for the equality $p_1^2 = p_2$ to be true the value of 'a' must be

59. If n be a positive integer such that $n \geq 3$, and the value of the sum to n terms of the series $1.n - \frac{n-1}{1!}(n-1) + \frac{(n-1)(n-2)}{2!}(n-2) - \frac{(n-1)(n-2)(n-3)}{3!}(n-3) + ...$ is 'S'. then the value of $\frac{2S}{19}$ is

Space for Rough work

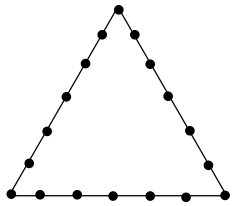
60. If the normals to the curve y = x^2 at the points P, Q and R pass through the point $(0, \frac{3}{2})$, then the radius of the circle circumscribing the triangle PQR is

61. The function *f* is one to one and the sum of all the intercepts of the graph is 5. The sum of all of the intercept of the graph of y = f^{-1}(x) is

62. In any triangle, if 'A' is the maximum value of $\frac{r^3}{r_1 r_2 r_3}$ then '54A' is equal to

63. The equation $a_1x^2 + a_2y^2 + z^2 + a_3xy + a_4yz + a_5zx + a_6x + a_7y + a_8z + a_9 = 0,$ where $a_i \in Z, i = 1,2,3,4,5,6,7,8,9$ is locus of the point which moves so that its distance from the line x=y=z is twice its distance from the plane x + y+z=1. Then the value of $\sum_{i=1}^{9} a_i$ is

64. If 18 points are arranged in triangular form as shown in figure. The number of triangles formed by these points is abc where abc is three digit number, a, b, c $\in$ z & a $\geq$ 1, b $\geq$ 0, c $\geq$ 0. The value of a + b + c is



65. Consider the determinant $\Delta(x) = \begin{vmatrix} (1+x)^{a_1b_1} & (1+x)^{a_1b_2} & (1+x)^{a_1b_3} \\ (1+x)^{a_2b_1} & (1+x)^{a_2b_2} & (1+x)^{a_2b_3} \\ (1+x)^{a_3b_1} & (1+x)^{a_3b_2} & (1+x)^{a_3b_3} \end{vmatrix}$ where a_1, a_2, a_3, b_1, b_2 & $b_3 \in$ N. Sum of coefficients of x and constant term of Δ(x) is

66. If $\lim_{x\to 0}\left(\cos x + p^3 \sin q^3 x\right)^{\frac{1}{x}} = e^{\frac{5}{2}}$, then the value of (pq) is

Space for Rough work

SECTION – D
(Numerical Based XXXXX.XX answer Type)

This section contains **3 questions**. Each question, when worked out will result in numerical answer Type with answer xxxxx.xx.

67. If $z^2 - z + 1 = 0$ then the value of $\sum_{i=1}^{24}(z_i + \frac{1}{z_i})^2$ is equal to

68. A sequence of equilateral triangle is drawn. The altitude of each is $\sqrt{3}$ times the altitude of the preceding traingle, the difference between the area of the first triangle and the sixth triangle is $968\sqrt{3}$ square unit. The perimeter of the first triangle is.

69. If $\Delta(a)$ is the area bounded by $xy^2 = 4a^2(2a - x)$ and its asymptotes then the value of $\Delta\left(S\sqrt{\frac{7}{22}}\right)$ is

Space for Rough work